Exp 8 · Effect of negative feedback (removing the emitter bypass capacitor)

Manual pp. 24–25 · LTspice · CO4, CO5 · the manual's two plots (≈ 42 dB and ≈ 4 dB) match the simulation below

5 BJT curves → 6 CE amplifier → 7 Emitter follower → **8 Feedback**

What you already have: from Exp 6, C_E bypasses R_E, and gain ≈ R_C/r_e. From Exp 7, a resistor in the emitter makes the output "follow" and pulls the gain down.

Two terms first:

- **Feedback:** feeding part of the output back to the input.
- **Negative** feedback opposes the input. It lowers the gain but steadies everything else.
- **Positive** feedback adds to the input. It leads to oscillation (Exp 11).
- **Current-series feedback:** the feedback voltage is produced by the output **current** ($I_E \approx I_C$ through R_E) and subtracted in **series** with the input ($V_{BE} = v_{in} - I_E R_E$).

MANUAL EXP 8, AIM

To observe the effect of negative feedback (emitter bypass capacitor) on the frequency response of an amplifier. Viva form: "Removing one capacitor cuts the gain from 122 to 1.7. Why would anyone want that?"

GUESS FIRST

Name one thing that gets **better** when the gain drops.

The idea

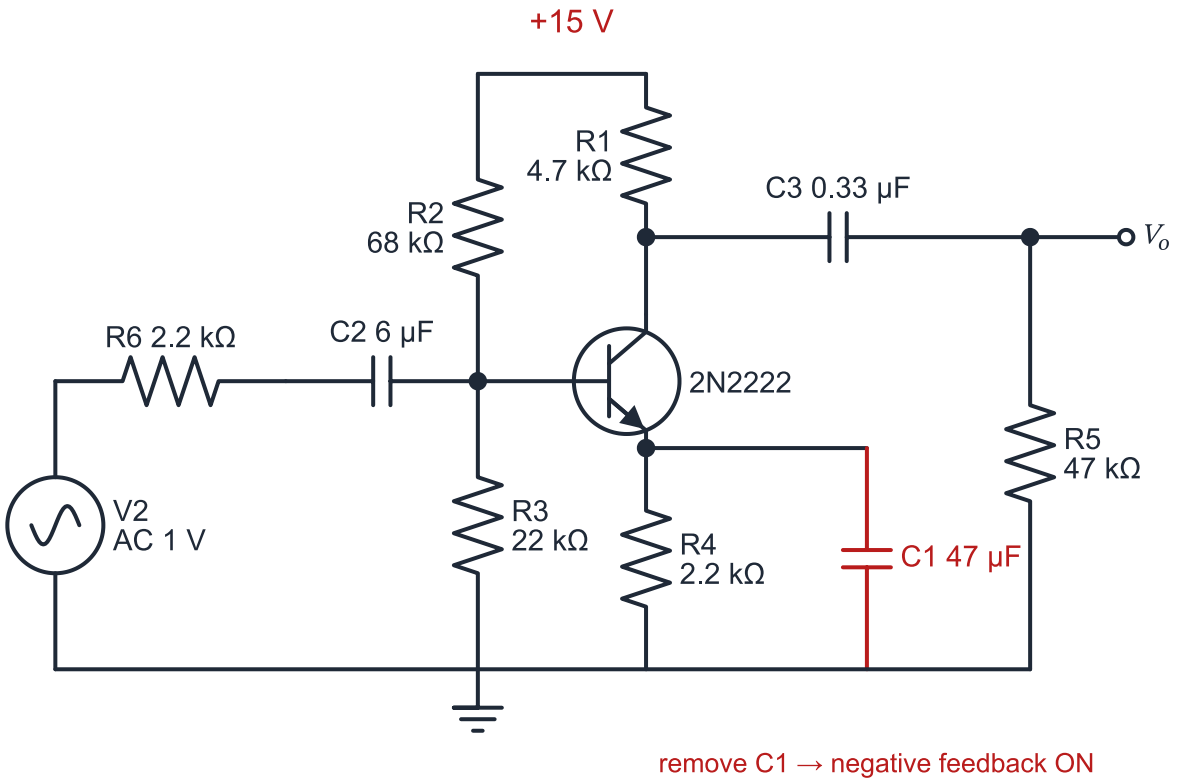


Fig 8.1, redrawn: a CE stage (2N2222) with load. C1 (red) is the bypass capacitor. Removing it is the whole experiment.

- **C1 in:** R4 is shorted for AC, so there is no feedback. $A_v \approx (R_C \parallel R_L)/r_e$, which is large.
- **C1 out:** the signal current through R4 (2.2 kΩ) makes a voltage that rises with the input, so the BE voltage the transistor actually sees shrinks. That is negative feedback. $A_v \approx (R_C \parallel R_L)/(r_e + R_E)$ = 4.27k/2.2k ≈ 1.9 at the base.

What negative feedback buys, at the cost of gain:

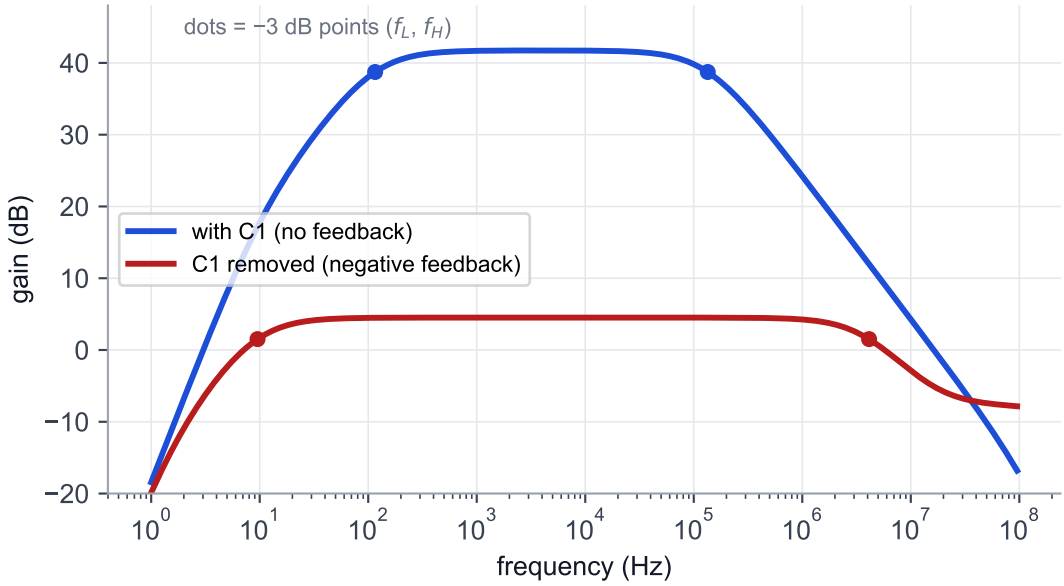
Property	Effect
gain	falls, but becomes **set by resistors** (R_C/R_E), not by β or temperature, so it is **stable**
bandwidth	**increases** (gain is traded for bandwidth)
distortion and noise	decrease
input resistance	increases (series feedback)

Do it

DO IT (LTSPICE)

1. Draw Fig 8.1. Q1 = **2N2222**. V2: AC amplitude `1`. Supply V1 = 15 V. Label the output `vo` (across R5, 47 kΩ).
2. Directive: `.ac dec 10 1 1Meg` (manual: 1 Hz – 1 MHz). Run. Plot V(vo): the solid line is the dB gain.
3. Read the peak dB with a cursor. Move the cursor to the frequencies where the gain is 3 dB lower: those are f_L and f_H.
4. **Delete C1** (or set it to `1p`). Run again and repeat.
5. Fill the table at 100 Hz, 500, 1k, 5k, 10k, 50k, 100k, 500k, 1M for both cases.

What you should see



LTspice. Blue: with C1, 41.7 dB peak, f_L 116 Hz, f_H 135 kHz. Red: without C1, 4.5 dB, f_L 9.5 Hz, f_H 4.1 MHz.

f (Hz)	100	500	1k	5k	10k	50k	100k	500k	1M
A_v with C1	79.8	119.1	121.4	122.1	121.9	114.5	98.0	31.7	16.3
A_{vf} without C1	1.67	1.68	1.68	1.68	1.68	1.68	1.68	1.67	1.63

	Without feedback (C1 in)	With feedback (C1 out)
maximum gain	122 (41.7 dB)	1.68 (4.5 dB)
f_L	116 Hz	9.5 Hz
f_H	135 kHz	4.1 MHz
bandwidth	≈ **135 kHz**	≈ **4.1 MHz**

Hand check (Q-point: V_B = 3.56 V, V_E = 2.90 V, so I_E = 1.32 mA and r_e = 19.7 Ω):

- R_{in} with feedback = 68k ∥ 22k ∥ β × 2.2k ≈ 16 kΩ. The source divider is 16/(16 + 2.2) = 0.88.
- A_{vf} ≈ 0.88 × 4.27k/2.22k = **1.70** (LTspice: 1.68) ✓.

WRITE IN YOUR RECORD

Result: bandwidth ≈ **135 kHz** without feedback (gain 41.7 dB). With negative feedback (C_E removed), the gain falls to 4.5 dB and the bandwidth rises to ≈ **4.1 MHz**: the higher cut-off is far above 1 MHz, as the manual expects.

Theory for the record: removing the emitter bypass capacitor leaves R_E unbypassed. The AC emitter current develops a voltage across R_E that subtracts from the input in series: current-series negative feedback. The voltage gain falls from ≈ R_C/r_e to ≈ $R_C/(r_e + R_E)$, but becomes stable (independent of β and temperature). The bandwidth increases, distortion and noise fall, and the input resistance rises. Negative feedback trades gain for bandwidth and stability.

CHECK 1

(a) Gain × bandwidth without feedback, and with feedback. Is the product the same? (b) Why is the 2N2222's own β almost irrelevant to the gain once C1 is out? (c) Which part of the response barely changed between the two cases at 100 kHz–1 MHz, and why?

Answers

Opening question. Negative feedback makes the gain **stable and predictable** (set by resistors, not β), **widens the bandwidth** (135 kHz → 4.1 MHz), and reduces distortion and noise. You add gain back with more stages; stability is hard to add later.

Check 1. (a) 122 × 135 kHz ≈ 16.5 MHz; 1.68 × 4.1 MHz ≈ 6.9 MHz. **Not the same**: "gain × bandwidth = constant" holds for a single dominant pole, not exactly for this circuit, but the direction (gain down, bandwidth up) holds. (b) $A_{vf} \approx (R_C \parallel R_L)/(r_e + R_E)$, and $R_E \gg r_e$, so β drops out. (c) The without-C1 curve is flat across it, while the with-C1 curve is already falling. With feedback, f_H moved out to MHz.